



Invited Commentary | Psychiatry

Precision Weight Loss Requires More Than Behavior Change

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Posttraumatic stress disorder (PTSD) is associated with increased risk of obesity. In a 2014 study using administrative data of 20 000 veterans,¹ those with PTSD lost less weight and were less likely to achieve clinically meaningful weight loss than veterans without mental health diagnoses in the Veterans Health Administration (VHA) weight-management program (MOVE!). Motivated by the study conclusion that MOVE! may require adaptation to better support individuals with mental health conditions, in this new study, Hoerster et al² describe a follow-up randomized clinical trial to test MOVE!+UP, a program designed to address unique barriers to weight loss among veterans with PTSD and elevated body mass index (BMI; calculated as weight in kilograms divided by height in meters squared). This trial enrolled 179 veterans with PTSD and overweight or obesity, assigning them to either standard MOVE! or an enhanced MOVE!+UP program that added cognitive behavioral therapy-based coping, relationship support, community engagement content; a therapeutic community walk; and individualized dietitian sessions. At 6 months, weight loss was similar between groups (mean [SD], 8.9 [13.7] lb in MOVE!+UP vs 7.8 [13.8] lb in MOVE!), with no significant difference after adjustment for starting weight and ethnicity (difference, -1.5 [95% CI, -5.9 to 2.9] lb; $P = .50$). Secondary outcomes, including PTSD symptoms and 12-month weight change, also did not differ. No adverse event-related withdrawals occurred, suggesting the intervention was safe but not more effective than standard MOVE!. What do these results say about the benefits of tailored weight loss interventions?

Although there may be an inclination to conclude from this study that tailored weight-loss programs are not worth the additional work, we feel this would be the wrong message. Both groups lost a clinically meaningful amount of weight (8-9 lb at 6 months, maintained through 12 months; 3%-4% of starting weight), and PTSD symptoms also improved in both groups. The 2 study groups may not have been sufficiently differentiated because the standard MOVE! program includes behavioral and cognitive elements that overlap with components emphasized in MOVE!+UP. This overlap may have reduced the contrast between conditions and limited the ability to detect incremental benefit attributable to the enhanced intervention. As such, the lack of significance observed for between-group differences may reflect the strong behavioral foundation inherent to MOVE!, rather than a lack of added value in the MOVE!+UP enhancements. However, MOVE!+UP likely was more resource intensive than MOVE!, and produced comparable benefit, suggesting that an economic evaluation may be warranted to help identify the most cost-effective strategy capable of delivering meaningful outcomes more broadly across the population.

In addition, expectations derived from the pilot randomized clinical trial may have been overly optimistic. The pilot study¹ reported a 14-lb reduction with a relatively small SD (3.7), whereas this trial by Hoerster et al² observed an 8.9-lb reduction but with a larger SD.² This discrepancy suggests that the current sample was considerably more heterogeneous; alternatively, that the pilot study sample was unusually homogeneous. In practical terms, these 2 studies may be capturing weight-loss responses under different conditions rather than simply reflecting sampling variability. If the goal is to replicate the magnitude of the pilot study's effect, a careful re-examination of how the pilot study participants differed from the cohort in this study by Hoerster et al² will be needed.

Many weight-loss trials show gradual weight regain after approximately 6 months.³ In this trial Hoerster et al,² weight loss achieved at 6 months appeared to be largely maintained through 12 months in both treatment groups. Some factors that have been found to improve weight loss include

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energy intake reduction (eg, through reducing unhealthy foods, portion control, decreasing food energy intake, increasing fruits and vegetables, and decreasing sugar-sweetened beverages), a high self-efficacy for weight management (eg, monitoring weight and tracking eating), and increased physical activity.⁴ Managing psychological stress is also important. Therefore, the sustained effect of both MOVE! groups suggests that the MOVE! platform itself is a successful approach to behavior change and provides evidence of intervention fidelity, as participants continued to maintain clinically meaningful weight reductions beyond the active treatment phase.

Although BMI of 25 or greater was an inclusion threshold for the trial by Hoerster et al,² there are limitations of applying a uniform weight-loss target across individuals with diverse body compositions.⁵ The mixed-effects models and extensive covariate adjustment strengthens internal validity, but these approaches do not fully address the inherently personalized nature of weight loss, particularly for individuals with severe obesity, for whom behavioral interventions alone may yield only modest benefit. In such cases, more intensive or multimodal strategies, including pharmacologic or surgical options, may be required to achieve clinically meaningful reduction of risks associated with obesity. Thus, interpreting weight-loss outcomes solely through the lens of mean group-level change may obscure the need for individualized treatment pathways that reflect patients' metabolic profiles, health risks, and feasible goals.

In addition, it is important to recognize that response to behavioral weight loss interventions is shaped by a wide range of psychological and behavioral factors and cannot be solely attributed to biology or genetics. Eating-related constructs (eg, emotional eating, disinhibition, restraint, perceived stress), age- and sex-related behavioral norms, and individual psychological characteristics (eg, motivation, self-efficacy, locus of control, and self-concept), as well as major life events, all play a role in determining who benefits most from behavioral approaches. Together, these multidimensional influences suggest that while behavioral interventions can be effective when executed well, their impact may be bounded for many individuals. This highlights the need for more nuanced, personalized treatment pathways that can identify individuals most likely to achieve meaningful benefit from behavioral interventions vs those who may require adjunctive or alternative therapies, such as glucagon-like peptide-1 receptor agonists.

Building on the need for more individualized treatment pathways, recent studies further support the value of personalized early-treatment prediction models.⁶ Rather than relying on fixed thresholds of percentage of weight lost, models that incorporate early weight-change patterns alongside age, sex, baseline weight, and deviations from expected weight have shown markedly better ability to identify individuals unlikely to achieve at least 5% weight loss by 1 year. In the POUNDS Lost trial,⁶ such models demonstrated increasing predictive accuracy across months 1, 2, and 3 (areas under the receiver-operating curve of 0.68, 0.75, and 0.79, respectively), whereas percentage of weight lost alone performed no better than chance. This suggests potential value in personalized decision-making that uses early response to guide whether a patient should continue behavioral treatment alone or transition to more intensive strategies.

The generalizability of these findings, as acknowledged by Hoerster et al,² is limited by the unique characteristics of veterans receiving care in the VHA system, who often have access to resources not available to the broader US population and whose PTSD presentations may differ from those seen in civilians. The sample was also restricted to veterans with overweight or obesity, and factors more common in this population (such as tobacco and alcohol use⁷) may influence weight loss outcomes but were not accounted for. Given that prior VHA weight-loss studies have shown minimal or modest effects, these contextual factors may explain why replication outside the VHA, or even within other veteran cohorts, may yield different results.

Nevertheless, findings from this randomized clinical trial demonstrate that meaningful and sustained weight loss is achievable in this population, and both interventions showed encouraging durability through 12 months. This consistency highlights the underlying strength of the MOVE! platform and reinforces the value of accessible, behavior-based weight-management strategies for veterans with PTSD.

ARTICLE INFORMATION

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